CSES — Bitwise Operations Problem Reference

All tasks: 1 s, 512 MiB.

1146 — Counting Bits

For all integers $1 \le v \le n$, count the total number of set bits in their binary representations.

In: `n`

Out: total number of 1 bits over $1, \ldots, n$

Constraints: $1 \le n \le 10^{15}$. Use 64-bit.

1655 — Maximum Xor Subarray

Given $x_1, \ldots, x_n$, find the maximum bitwise XOR of any nonempty contiguous subarray.

In: `n`; `x_1 ... x_n`

Out: maximum subarray XOR

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3191 — Maximum Xor Subset

Choose any subset of the array and XOR its elements. Find the maximum obtainable XOR value.

In: `n`; `x_1 ... x_n`

Out: maximum subset XOR

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3211 — Number of Subset Xors

Consider XOR values of all subsets, including the empty subset with XOR 0. Count how many distinct values are obtainable.

In: `n`; `x_1 ... x_n`

Out: number of distinct subset XORs

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3192 — K Subset Xors

Consider the XOR of every one of the 2^n subsets, including the empty subset. Values are counted with multiplicity. Print the k smallest values in nondecreasing order.

In: `n k`; `x_1 ... x_n`

Out: k smallest subset XORs, including duplicates

Constraints: $n \le 2 \cdot 10^5$, $k \le \min(2^n, 2 \cdot 10^5)$, $0 \le x_i \le 10^9$.

3233 — All Subarray Xors

Find every distinct integer that occurs as the XOR of some nonempty contiguous subarray.

In: `n`; `x_1 ... x_n`

Out: first count k, then the k XOR values in increasing order

Constraints: $n \le 2 \cdot 10^5$, $0 \le x_i \le 10^6$.

2419 — Xor Pyramid Peak

The bottom row is $a_1, \ldots, a_n$. Every cell above is XOR of its lower-left and lower-right cells. Find the single top cell.

In: `n`; `a_1 ... a_n`

Out: topmost value

Constraints: $n \le 2 \cdot 10^5$, $1 \le a_i \le 10^9$.

3194 — Xor Pyramid Diagonal

Build the same XOR pyramid. Print the *leftmost* value of every row, ordered from the bottom row up to the top.

In: `n`; `a_1 ... a_n`

Out: n leftmost row values, bottom $\to$ top

Constraints: $n \le 2 \cdot 10^5$, $1 \le a_i \le 10^9$.

3195 — Xor Pyramid Row

Build the XOR pyramid from the given bottom row. Print all values on the kth row *from the top*; that row has exactly k entries.

In: `n k`; `a_1 ... a_n`

Out: the k values on row k from the top

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le a_i \le 10^9$.

1654 — SOS Bit Problem

For every input element x, count array elements y satisfying each condition:

- $x \mid y = x$;
- $x \,\&\, y = x$;
- $x \,\&\, y \ne 0$.

Counts include occurrences, so equal array values are counted separately.

In: `n`; `x_1 ... x_n`

Out: three counts for each input element, in input order

Constraints: $n \le 2 \cdot 10^5$, $1 \le x_i \le 10^6$.

3141 — And Subset Count

For every $k = 0, 1, \ldots, n$, count the nonempty subsets whose bitwise AND of all chosen elements is exactly k.

In: `n`; `a_1 ... a_n`

Out: $n + 1$ counts for $k = 0, \ldots, n$, modulo $10^9 + 7$

Constraints: $n \le 2 \cdot 10^5$, $0 \le a_i \le n$.